A Hybrid Reinforcement Learning Framework for Hard-Latency Constrained Resource Scheduling

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***Abstract*—In the forthcoming sixth generation (6G) era, extend reality (XR) has been regarded as an emerging application for ultrareliable and low-latency communications (URLLC) with new traffic characteristics and more stringent requirements. In addition to the quasi-periodical traffic in XR, burst traffic with both large frame size and random arrivals in some real world low-latency communication scenarios has become the leading cause of network congestion or even collapse, and there still lacks an efficient algorithm for the resource scheduling problem under burst traffic with hard latency constraints. We propose a novel hybrid reinforcement learning framework for resource scheduling with hard-latency constraints (HRL-RSHLC), which reuses polices from both old policies learned under other similar environments and domain-knowledge-based (DK) policies constructed using expert knowledge to improve the performance. The joint optimization of the policy reuse probabilities and new policy is formulated as an Markov decision problem (MDP), which maximizes the hard-latency constrained effective throughput (HLC-ET) of users. We prove that the proposed HRL-RSHLC can converge to KKT points with an arbitrary initial point. Simulations show that HRL-RSHLC can achieve superior performance with faster convergence speed compared to baseline algorithms.**

***Index Terms*—Burst traffic, hard-latency constraints, multiuser multiple-input–multiple-output (MU-MIMO), reinforcement learning (RL), resource scheduling.**

I. Introduction

A. Background

ULTRARELIABLE and low-latency communications (URLLC), which is one of the major communication scenarios of the fifth generation (5G) wireless communication networks, has always been a key requirement for many applications, such as public safety, telemedicine and etc [1], [2], [3]. Average latencies are not of interest for URLLC applications, as an instantaneous interruption in the transmitted data will lead to a poor performance of the overall system. Meeting hard delay constraints plays a critical role in supporting URLLC and existing works have proposed various methods to meet the delay constraints. For example, a novel framework utilizing a risk-constraint deep reinforcement learning (DRL) algorithm was proposed in [4] to improve the scheduling performance by limiting the expectation of quality of service (QoS) to a threshold. Huang and Kadoch [5] proposed a spectrum resource scheduling strategy based on the reinforcement learning (RL) approach to meet the low latency constraints.

However, previous research efforts on the conventional URLLC use cases in 5G only support short packets transmission, which cannot meet all requirements of the future wireless communications. Researchers start to focus on the sixth generation (6G) wireless communication networks, and extend reality (XR), which is an umbrella term for different types of realities, such as virtual reality (VR), augmented reality (AR), and mixed reality (MR), has been regarded as an emerging application for URLLC in 6G with new traffic characteristics and more stringent requirements. Different from short-packet transmissions in conventional URLLC, XR frame has a much larger size and requires multiple timeslots to complete the transmission, which makes resource scheduling more difficult to meet the hard delay constraints [6]. To overcome this challenge, Chaccour et al. [7] analytically derived the end-to-end delay distribution over a Terahertz (THz) link, taking into account the VR frame's processing and transmission delays. Chen et al. [8] considered a cloud XR architecture, formulated a resource allocation problem to maximize the number of satisfied users under the data rate, reliability and latency requirements for each user, and solved the problem approximately by jointly considering the user admission control and a frame-level integrated transmission problem.

In addition to the quasi-periodical traffic in XR, burst traffic with both large-frame size and random arrivals in some real world low latency communication scenarios has become the leading cause of network congestion or even collapse [9]. However, to the best of our knowledge, existing works in the literature have not considered resource scheduling with hard delay constraints under burst traffic, which still face various technical challenges to be addressed.

B. Related Works

There are two major methods to solve resource scheduling problems with delay constraints.

1) Non-RL Methods: One common approach to deal with scheduling problems with latency constraints is to assume

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simple and known traffic/channel statistics. Kharel et al. [10] assumed the distribution of the traffic and channel was known, then attained accurate closed-form expressions for the outage probability, and proposed a mini-slots-based scheduling framework to serve URLLC users under latency deadlines. A hybrid multiple access (HMA) solution was proposed in [11] based on the outage probability for URLLC traffic. In [12], it was assumed that all the packets are of equal size and have a strict delay constraint, and the objective of the scheduler is to minimize the total transmit power subject to strict delay constraints and the author developed the upper and lower bound on output rates and developed a two step solution. Reference [13] considered a multiuser multiple-input–multiple-output (MU-MIMO) with delay constraints, in which a certain size of data in each user's queue needs to be transmitted within the deadline. Hassan and Assaad [14] considered a downlink OFDMA system with at most one packet in each user's queue, in which all the packets cannot be delivered before their deadline would be dropped, formulated a constrained resource optimization problem and then replaced the reverse convex constraint by a linear constraint, transforming the original problem into a convex optimization problem, and achieved the delay constraints for each packet. [12], [13], and [14] all assume known traffic/channel statistics, which is difficult to achieve in real world application scenarios with burst traffic and dynamic wireless environment.

2) RL Methods: RL method is a common approach to get rid of the unrealistic assumptions on traffic/channel statistics. Kasgari and Saad [15] proposed a deep-RL framework to jointly allocate resource blocks (RBs) and power, and dynamically measures the end to end reliability and the delay of each user, achieving the low latency constraints. Saggese et al. [16] proposed a DRL approach to train an agent acting as a scheduler and never violate URLLC latency requirements. [17] formulated the scheduling problem with a strict delay constraint on each queued packet stored in the buffer as a Markov decision problem (MDP) and solved it using relative value iteration (VI) algorithm. To obtain the optimal tradeoff between delay and power consumption for a given power constraint in a communication system whose traffic/channel conditions can change over time, Zhao and Lee [18] formulated the problem as an infinite-horizon MDP and then Q-learning was adopted to solve this problem. Shafieirad and Adve [19] proposed the use of RL and deep learning to address the max delay latency constraint through a sequential decision making model. However, the above existing works on RL-based resource scheduling focus on conventional URLLC or other simple scenarios, and have not considered both hard delay constraints and burst traffic.

C. Contributions

The above existing works in the literature have not fully addressed the problem of resource scheduling with hard delay constraints under burst traffic. Although there have been some attempts to apply RL for this problem, the large state/action space, the hard latency constraints and the more unpredictable burst traffic with multitimeslots transmissions make the conventional RL algorithms hard to converge. To solve this challenging problem, we propose a novel resource scheduling algorithm called hybrid RL framework for resource scheduling with hard-latency constraints (HRL-RSHLC) in this article. The main contributions of this work are,

1) *HRL-RSHLC:* Unlike the conventional RL algorithms which aim to optimize a single policy, we propose a hybrid RL framework consists of a mixture of a deep-neural network (DNN) parameterized policy (which is called the new policy), a domain-knowledge-based (DK) policy constructed using expert knowledge (which is called the DK policy) and old policies trained under other similar environments. The joint optimization of the probabilities for using each policy and DNN parameters of the new policy are formulated as an MDP, where the objective is to maximize the hard-latency constrained effective throughput (HLC-ET) of users. In particular, the hard delay constraints are embodied in the objective function, that is, only packets which have been successfully delivered before the hard delay constraints would be considered in the HLC-ET, which avoid the use of constrained MDP (CMDP). However, due to the unpredictable burst traffic with multitimeslots transmissions, the implicit hard delay constraints in the objective function also brings sparse rewards, which poses challenges for fast-convergent algorithm design.
2) *A Fast Convergent Resource Scheduling Algorithm Based on Hybrid RL to Maximize the HLC-ET:* Based on the above hybrid RL framework, we propose a novel resource scheduling algorithm called HRL-RSHLC, to solve the MDP of maximizing the HLC-ET. In HRL-RSHLC, both policy/data reuse as well as the domain specific knowledge are exploited to accelerate the convergence speed. Specifically, instead of directly controlling the user scheduling and multiple-input–multiple-output (MIMO) precoder, the HRL-RSHLC only controls a priority weight vector, and the user scheduling and MIMO precoder are indirectly determined by maximizing the weighted sum-rate (WSR) using a classic iterative algorithm. Such a design can significantly reduce the action space. The optimization of policy reuse probabilities can further accelerate the convergence speed. If an old policy performs well in the current environment, its reuse probability will automatically be increased by the hybrid RL algorithm to accelerate the initial convergence speed. Moreover, we introduce a DK policy in which the action (weight vector) is chosen as Q-weighted greedy scheduling algorithm [20]. The DK policy is shown to perform well for light to moderate traffic loading, and can provide a stable scheduling performance. As such, even if all the DNN-based policies cannot work well (e.g., due to the limited representational capacity of practical DNNs and the lack of interpretability), the hybrid RL algorithm will automatically increase the reuse probability of the DK policy to avoid any catastrophic failure of DNN-based policies. Finally, HRL-RSHLC updates the policy using both stored old experiences and newly added

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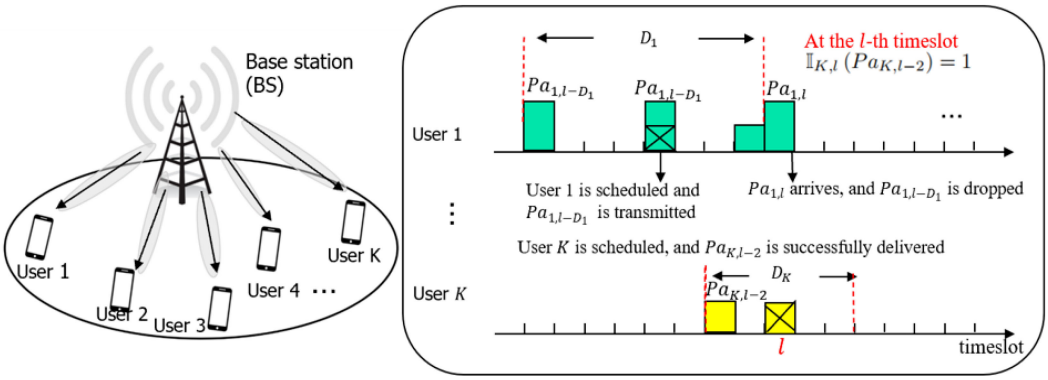

Fig. 1. System model.

data, and alleviates the issues caused by the sparsity of reward.

3) *Convergence Analysis and Theoretical Performance Guarantee:* We prove that the proposed HRL-RSHLC algorithm can converge to KKT points with an arbitrary initial point. Simulations show that HRL-RSHLC can achieve superior performance with faster convergence speed compared to baseline algorithms.

The rest of this article is organized as follows. In Section II, we illustrate the system model considered in this article. In Section III, we elaborate the hybrid RL framework and problem formulation for burst traffic with hard latency constraints. In Section IV, we propose a HRL-RSHLC algorithm to solve the formulated resource scheduling problem. In Section V, we prove that the proposed HRL-RSHLC can converge to KKT points with an arbitrary initial point. Section VI showcases the simulation results. Finally, we conclude this article in Section VII.

II. System Model

We consider a downlink MU-MIMO system under burst traffic. As shown in Fig. 1, a base station (BS) serves a set of K users $\mathcal{K}$. The BS is equipped with N_T antennas and each user is equipped with a single antenna. At the BS, each user is assigned an individual buffer. Data packets from some higher-layer applications randomly arrive at the buffers and are stored in the form of queues and the BS dynamically schedules the resource for the downlink transmission of each data queue.

A. Downlink Multiuser MIMO Transmission

The proposed algorithm is efficient for any precoding scheme and power allocation scheme. In the simulations, we adopt regularized zero-forcing (RZF) and equal power allocation under the single-stream transmission mode, which are widely applied in real systems.

We suppose that time is divided into time slots of duration τ with index l. Let pow_i and $\boldsymbol{v}_i \in \mathbb{C}^{N_T\times 1}$ denote the transmit power and normalized precoder for user i. At the lth time slot, the received signal at user i can be formulated as

$$\boldsymbol{y}_i = \boldsymbol{h}_i\sqrt{\text{pow}_i}\boldsymbol{v}_i d_i + \sum_{j\in\mathcal{B}/i}\boldsymbol{h}_i\sqrt{\text{pow}_j}\boldsymbol{v}_j d_j + \boldsymbol{n}_i \tag{1}$$

where $\boldsymbol{h}_i \in \mathbb{C}^{1\times N_T}$ is the channel vector of user i, the time slot index l is omitted for conciseness, and $\mathcal{B}$ denotes the scheduled user set, which will be given in detail in Section III-A. d_i is the normalized data symbols, i.e., $\mathbb{E}[|d_i|^2] = 1$, and $\boldsymbol{n}_i \sim \mathcal{CN}(0, \sigma_i^2)$ is additive white Gaussian noise (AWGN) with variance σ_i^2. Therefore, the signal-to-interference-plus-noise-ratio (SINR) of user i can be formulated as

$$\text{SINR}_i = \frac{\text{pow}_i|\boldsymbol{h}_i^H\boldsymbol{v}_i|^2}{\sum_{j\in\mathcal{B}/i}\text{pow}_j|\boldsymbol{h}_i^H\boldsymbol{v}_j|^2 + \sigma_i^2} \tag{2}$$

then the data rate of ith user is given by

$$R_i = B\log_2\left(1 + \frac{\text{pow}_i|\boldsymbol{h}_i^H\boldsymbol{v}_i|^2}{\sum_{j\in\mathcal{B}/i}\text{pow}_j|\boldsymbol{h}_i^H\boldsymbol{v}_j|^2 + \sigma_i^2}\right) \quad \forall i \tag{3}$$

where B is the bandwidth, and $\boldsymbol{V} = [\boldsymbol{v}_i]_{i\in\mathcal{K}} \in \mathbb{C}^{N_T\times|\mathcal{B}|}$ is the normalized RZF precoding matrix, which is given by

$$\boldsymbol{V} = \boldsymbol{H}_B^H\left(\boldsymbol{H}_B\boldsymbol{H}_B^H + \alpha\boldsymbol{I}\right)^{-1}\boldsymbol{\lambda}^{\frac{1}{2}} \tag{4}$$

where α is a regularization factor, $\boldsymbol{\lambda}$ is a diagonal matrix for normalization, and $\boldsymbol{H}_B(t) \in \mathbb{C}^{|\mathcal{B}|\times N_T}$ is channel matrix formed by merging the equivalent channels of scheduled users.

Remark 1: For clarity, we assume that the users have a single-antenna and the BS has perfect channel state information (CSI). As will be discussed in Section III-A, the proposed resource allocation algorithm does not directly control the MIMO precoder but only control the priority weight of each user. As such, the proposed algorithm does not rely on any specific MU-MIMO transmission scheme at the physical layer, and it can be directly applied in a system when the users have multiple antennas or the BS has imperfect CSI. For example, when the BS has imperfect CSI, the precoder can be calculated using the estimated channel, i.e., in (4), $\boldsymbol{H}_B$ is replaced by the estimated channel $\hat{\boldsymbol{H}}_B$. If the users have multiple antennas, we can simply use the block diagonal (BD) precoder or any other precoder designed for the case with multiantenna users.

The key notations are summarized in Table I.

B. Traffic and Queue Dynamic Model

We assume that the data packets only arrive at the start of each timeslot. Specifically, at the lth timeslot, a single data

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TABLE I
SUMMARY OF NOTATIONS

$K/\mathcal{K}$	Number of users/ User set
N_T/N_R	Number of BS/user antenna
$\boldsymbol{h}_i$	Channel of user i
l	Time slot index
$pow_i/\boldsymbol{v}_i$	Power/precoder of user i
$Pa_{i,l}$	Packet that arrives at user i
$\bar{Q}_{i,l}$	Original length of $Pa_{i,l}$
$Q_{i,l}$	Remaining length of $Pa_{i,l}$
PA	Packet arrival probability
D_i	Delay constraint for user i
R_i	Data rate of user i
p_n	Probability to use policy n
τ	Time slot duration

packet $Pa_{i,l}$ of length $\bar{Q}_{i,l}$ Kbit arrives at the queue of user i with a probability PA. The length of arrived data is random with $\mathbb{E}(\bar{Q}_{i,l}) = \lambda_i$.

The delay constraint for user i is D_i timeslots, which means that if a packet arrives at user i's queue at the lth timeslot, and at the $(l+D_i)$th timeslot it has not been successfully delivered, then it would be dropped out of the queue at this timeslot. Apparently, there are at most D_i packets in the queue of user i. To better capture the state of each packet in the queue, we denote $Q_{i,l}$ as the remaining data size of the packet $P_{i,l}$, and $B(Pa_{i,l}) = \sum_{l'=1}^{D_i-1} Q_{i,l-l'}$ as the length of the packet backlog in front of $Pa_{i,l}$. The arrived data packets are served according to the first-come-first-served (FCFS) protocol. In the FCFS protocol, packets are serviced in the order they arrive, that is, the first packet to arrive is the first one to be transmitted, followed by the next one in queue, and so on. $Q_{i,l}$ is the remaining size of packet $Pa_{i,l}$ and $B(Pa_{i,l}) = \sum_{l'=1}^{D_i-1} Q_{i,l-l'}$ is the length of the packet backlog in front of $Pa_{i,l}$. Thus, according to the FCFS protocol, the packet $Pa_{i,l}$ will not be served until $B(Pa_{i,l}) = 0$. In hard-latency constrained transmissions, we focus on the following two crucial cases.

1) *Data Being Dropped:* Packets failed to be delivered before their deadlines would be dropped. Specifically, at the lth time slot, the packet $Pa_{i,l-D_i}$in the queue of user i would be dropped if $Q_{i,l-D_i} > R_i(l-1)\tau$.
2) *Data Being Successfully Delivered:* We define a binary functions $\mathbb{I}_{i,l}(Pa_{i,l'})\quad \forall i, l$, to indicate whether the packet $Pa_{i,l'}$ is successfully delivered at the lth time slot

$$\mathbb{I}_{i,l}\big(Pa_{i,l'}\big) = \begin{cases} 1, & \text{if } B\big(Pa_{i,l'}\big) + Q_{i,l'} \le R_i(l)\tau \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

For ease of understanding, we show a possible state of the data queues in Fig. 1. As shown in Fig. 1, packet $Pa_{1,l-D_1}$ arrived at user 1's queue D_1 timeslots ago, then user 1 was scheduled 4 timeslots ago and part of the packet was transmitted, which is the green section marked with an X in the figure. At the lth timeslot, packet $Pa_{1,l-D_1}$ has not been successfully delivered $(R_1(l-1)\tau < Q_{1,l-D_1})$ so it is *dropped* and a new packet $Pa_{1,l}$ arrives at user 1's queue at the same time. At the lth timeslot, user K is scheduled and its transmission rate $R_K(l)\tau \ge Q_{K,l-2}$, so $Pa_{K,l-2}$ is *successfully delivered.*

C. Hard-Latency Constrained Effective Throughput

In this article, we focus on optimizing the HLC-ET of users. Specifically, at the lth time slot, the instantaneous HLC-ET of users can be defined as

$$\frac{1}{\tau}\sum_{i=1}^{K}\sum_{l'=0}^{D_i-1} \mathbb{I}_{i,l}\big(Pa_{i,l-l'}\big)\bar{Q}_{i,l-l'} \tag{6}$$

where τ is the duration of each timeslot, $Pa_{i,l-l'}$ is the packet that arrived at user i's queue at $l-l'$th timeslot, and $\bar{Q}_{i,l-l'}$ is the original length of packet $Pa_{i,l-l'}$. The delay constraint of user i is D_i, so at the lth timeslot, the oldest packet in user i's queue is $Pa_{i,l-(D_i-1)}$. The indication function $\mathbb{I}_{i,l}(Pa_{i,l-l'}) = 1$ means that $Pa_{i,l-l'}$ with the original packet length $\bar{Q}_{i,l-l'}$ is successfully delivered at the lth timeslot, which contributes to the HLC-ET with the term $(1/\tau)\bar{Q}_{i,l-l'}$. Without loss of generality, we set $\tau = 1$. When packet $Pa_{i,l-l'}$ is delivered successfully at the lth time slot, the original data size $\bar{Q}_{i,l-l'}$ would be included in the HLC-ET. To simplify the presentation, we define vectors $\boldsymbol{Q}(l) = [Q_{1,l-D_1+1}, \ldots, Q_{1,l}, \ldots, Q_{K,l-D_K+1}, \ldots, Q_{K,l}]^T \in \mathbb{R}^{\sum_{i=1}^{K} D_i}$ and $\bar{\boldsymbol{Q}}(l) = [\bar{Q}_{1,l-D_1+1}, \ldots, \bar{Q}_{1,l}, \ldots, \bar{Q}_{K,l-D_K+1}, \ldots, \bar{Q}_{K,l}]^T \in \mathbb{R}^{\sum_{i=1}^{K} D_i}$.

III. HYBRID RL FRAMEWORK AND PROBLEM FORMULATION

In this section, we propose a hybrid RL framework to achieve efficiently online scheduling in the considered system. First, we explain the motivation of choosing the priority weight as the action of the agent. Then, the problem formulation based on MDP is presented. Finally, we introduce the details of the hybrid RL framework.

A. Resource Scheduling Based on Weighted Sum-Rate Maximization

Some previous works applying the RL algorithms to solve resource scheduling problems have chosen discrete actions, e.g., the scheduling algorithm proposed in [17] chose the number of transmitted packets as the action. However, since the values of the states, e.g., the CSI, are continuous, applying discrete actions may decrease the performance of RL. On the other hand, the action space of discrete actions would be large when MU scheduling is considered, extremely when the number of users is large.

In this article, we design a resource scheduling algorithm based on controlling the priority weights of users, and the user scheduling and MIMO precoder/power allocation are indirectly determined by maximizing the WSR using a classic iterative algorithm. Such a design can significantly reduce the action space and speed up the convergence compared to directly controlling all of the variables, i.e., the user scheduling and power allocation. When the data rate region is strongly convex, using the priority weights as the control action and maximizing the WSR will not lose any optimality, as explained

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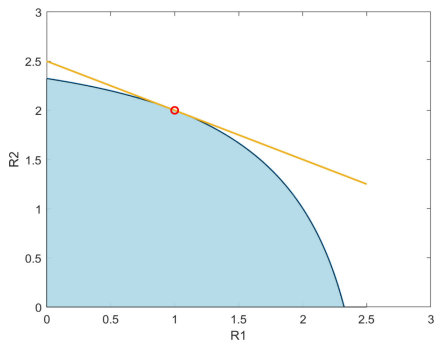

Fig. 2. Illustration of a strongly convex rate region.

below. For a given weight, the unique WSR maximization rate point is the tangent point of the plane determined by the weight and the rate region, as shown in Fig. 2. This means that any Pareto rate point on the boundary of the strongly convex rate region can be achieved by maximizing the WSR with a proper weight vector. Since the optimal user scheduling and power allocation must also achieve a certain Pareto rate point, and we can always achieve the same Pareto rate point by controlling the weight vector, directly controlling the weight vector will not lose any optimality.

It is well-known that the capacity region of Gaussian MIMO broadcast channel (BC) is strongly convex under a total power constraint. Thus, directly controlling the priority weights will not lose any optimality for capacity achieving physical layer schemes (such as dirty paper coding [21]). Simulations show that directly controlling the priority weights is still very efficient for resource allocation under other suboptimal but more practical physical layer scheme, such as RZF beamforming [20], even when the rate region is not strongly convex in this case. Indeed, we present simulation results in Section VI that compares the performance of choosing the weight as action and directly controlling the user scheduling and power allocation in the action. Simulation results show that reducing the action space can significantly improve the convergence speed. In fact, HRL-RSHLC controlling all actions failed to converge to a good solution, which is probably due to fact that an enlarged action space makes it more likely to get stuck in a bad local optimum.

We define the priority weight vector as $\boldsymbol{w} = [w_1, \ldots, w_K]$, then in the proposed design, the optimal user scheduling and power allocation scheme is obtained by maximizing the WSR of users. The optimization problem can be formulated as

$$\max_{\mathcal{B}, \{\text{pow}_i\}_{i\in\mathcal{B}}} \sum_{i\in\mathcal{B}} w_i \cdot B \log_2\left(1 + \frac{\text{pow}_i |\boldsymbol{h}_i^H \boldsymbol{v}_i|^2}{\sum_{j\in\mathcal{B}/i} \text{pow}_j |\boldsymbol{h}_i^H \boldsymbol{v}_j|^2 + \sigma_i^2}\right) \tag{7}$$

where $\mathcal{B}$ denotes the scheduled user set, $\{\text{pow}_i\}_{i\in\mathcal{B}}$ is determined according to equal power allocation scheme, and $\text{pow}_i = 0 \quad \forall i \notin \mathcal{B}$. Note that the proposed hybrid RL framework works for any resource allocation policy that aims at maximizing the WSR, and we adopt the greedy user selection algorithm with RZF MIMO precoder and equal power allocation [20] in this article due to its wide application in practical systems. We assume that the BS knows the CSI of all users and the greedy user selection algorithm with RZF MIMO precoder is based on the known CSI at the BS. Specifically, the greedy user scheduling algorithm selects users by round: In each round, it finds one user that be added to the selected user set to maximize the WSR, until no more users can be found that would increase the WSR. Clearly, the proposed RL framework based on controlling priority weights works for any other MU-MIMO transmission scheme, as long as one can design an iterative algorithm to solve the WSR maximization problem.

B. Problem Formulation Based on MDP

An MDP is denoted as a tuple $(\mathcal{S}, \mathcal{A}, R, P)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function. $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the transition probability function, and $P(\boldsymbol{s}'|\boldsymbol{s}, \boldsymbol{a})$ denotes the transition probability from state s to state s' under action $\boldsymbol{a}$. A policy $\pi : \mathcal{S} \to \mathbf{P}(\mathcal{A})$ is a map from states to probability distributions over actions, and $\pi(\boldsymbol{a}|\boldsymbol{s})$ denotes the probability of choosing action $\boldsymbol{a}$ in state $\boldsymbol{s}$. Due to the curse of dimensionality, modern RL algorithms, e.g., DRL-based algorithms, usually parameterize the policy by function approximations with high representation capability, e.g., DNN. In this article, we denote $\pi_{\boldsymbol{\theta}}$ as the policy parameterized by $\boldsymbol{\theta}$, as will be detailed later in Section III-C.

1) *The State Space $\mathcal{S}$:* $\mathcal{S}$ is a composite space consisting of the queue state space and the channel state space, i.e., the current state information at the lth time slot is denoted as $\boldsymbol{s}_l = \{\boldsymbol{Q}(l), \bar{\boldsymbol{Q}}(l), \boldsymbol{H}(l)\}$, where $\boldsymbol{H}(t) \in \mathbb{C}^{K\times N_T}$ is channel matrix formed by merging the equivalent channels of all users.
2) *The Action Space $\mathcal{A}$:* the priority weight vector space constitute the action space $\mathcal{A}$, i.e., the action at the lth time slot is $\boldsymbol{a}_l = \{\boldsymbol{w}_l\}$. Specifically, the action $\boldsymbol{a}_l$ is sampled according to a policy $\pi_{\boldsymbol{\theta}} : \mathcal{S} \to \mathbf{P}(\mathcal{A})$.
3) *The Transition Probability Function P:* the function $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is an unknown transition probability function related to the statistics of the unknown statistics of environment model, where $P(\boldsymbol{s}_{l+1}|\boldsymbol{s}_l, \boldsymbol{a}_l)$ denotes the probability of transition to state $\boldsymbol{s}_{l+1}$ from state $\boldsymbol{s}_l \in \mathcal{S}$ with an action $\boldsymbol{a}_l$. The transition probability P and policy $\pi_{\boldsymbol{\theta}}$ together determine the probability distribution of the trajectory $\{\boldsymbol{s}_0, \boldsymbol{a}_0, \boldsymbol{s}_1, \ldots\}$.
4) *The Reward Function R:* at each timeslot l, the instantaneous HLC-ET of users is set to be the reward. The greedy user scheduling algorithm is applied to maximize the WSR based on $\boldsymbol{a}_l$, and we have $R(\boldsymbol{s}_l, \boldsymbol{a}_l) = \sum_{i=1}^{K} \sum_{l'=0}^{D_i-1} \mathbb{I}_{i,l}(Pa_{i,l-l'}) \bar{Q}_{i,l-l'}$.

In this article, we consider the problem of maximizing the average HLC-ET, which can be formulated as an MDP

$$\min_{\boldsymbol{\theta}\in\Theta} J(\theta) \triangleq \lim_{L\to\infty} \frac{1}{L} \mathbb{E}_{p_s\sim\pi_{\boldsymbol{\theta}}}\left[-\sum_{l=0}^{L-1}\sum_{i=1}^{K}\sum_{l'=0}^{D_i-1} \mathbb{I}_{i,l}\big(Pa_{i,l-l'}\big) \cdot \bar{Q}_{i,l-l'}\right] \tag{8}$$

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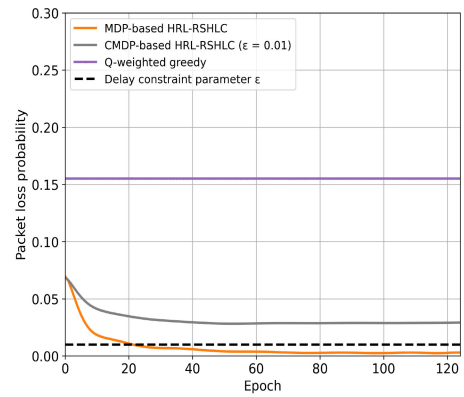
where $p_s \sim \pi_{\boldsymbol{\theta}}$ denote the probability distribution of the trajectory under policy $\pi_{\boldsymbol{\theta}}$. Note that there is no need to add an explicit constraint for the probability of violating the hard delay constraint due to the following reason. When all of the packets have the same size and delay constraint $D_{\max}$, the average HLC-ET is equal to the product of the packet arrival rate and the successful transmission probability, i.e., $A * (1 - \Pr(D > D_{\max}))$, where A is the packet arrival rate. Therefore, maximizing the HLC-ET is equivalent to minimizing the probability of violating the hard delay constraint $\Pr(D > D_{\max})$.

Another possible formulation is to maximize the average throughput with an explicit constraint for the probability of violating the hard delay constraint, i.e., considering the following constrained MDP (CMDP)

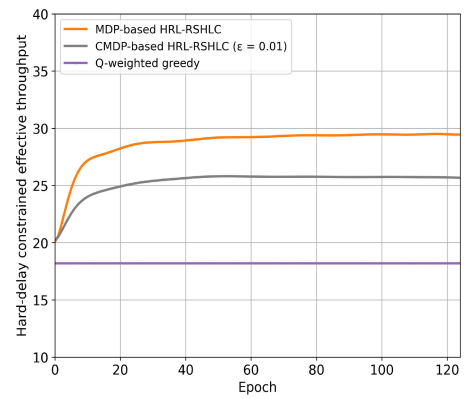
$$\min_{\boldsymbol{\theta}\in\Theta} J(\theta) \triangleq \lim_{L\to\infty} \frac{1}{L}\mathbb{E}_{p_s\sim\pi_{\boldsymbol{\theta}}}\left[-\sum_{l=0}^{L-1}\sum_{i=1}^{K}\mathrm{THP}_i(l)\right]$$
$$\text{s.t.} \lim_{L\to\infty} \frac{1}{L}\mathbb{E}_{p_s\sim\pi_{\boldsymbol{\theta}}}\left[\sum_{l=0}^{L-1}\frac{u'(l)}{u(l)}\right] - \epsilon \le 0 \tag{9}$$

where $\mathrm{THP}_i(l) = \min(R_i(l), \sum_{j=1}^{D_i} Q_{l-j+1})$ represents the throughput of user i, $u'(l)$ is the number of packets transmitted to users in timeslot l, whose latency exceeds $D_{\max}$, $u(l)$ is the total number transmitted to users, and ϵ is the maximum allowable probability of violating the hard delay constraint. $(u'(l)/u(l))$ is the estimated value of $(1 - \Pr(D > D_{\max}))$ at the lth timeslot [15].

In Fig. 3(a) and (b), we compare the probability of ensuring hard latency for the HRL-RSHLC algorithms based on the MDP and CMDP formulations, respectively. It is observed from the simulations that solving the CMDP problem requires higher computational complexity. Moreover, the MDP-based HRL-RSHLC can achieve a higher effective throughput because it is designed to directly maximize the effective throughput and it also has a better convergence behavior. Thus, in this article, we shall design the HRL-RSHLC algorithm based on the MDP formulation in (8).



(a)



(b)

Fig. 3. Comparison with CMDP-based HRL-RSHLC. (a) Comparison of packet loss probability. (b) Comparison of HLC-ET.

C. Hybrid RL Framework

To accelerate the convergence speed and reduce the interaction costs with the environment in RL, various methods have been proposed in the literature and most of them consider transferring learning (TL) or domain knowledge exploitation, which are heuristic without rigorous convergence analysis or theoretical performance guarantee, e.g., the probability of using each policy is determined by a heuristic method and the convergence of the overall algorithm is not guaranteed in the policy reuse method [22]. Since the proposed hybrid RL framework is inspired by the policy reuse method, we shall first review this method as preliminary and motivation for the proposed framework.

1) Preliminary on Policy Reuse: Policy reuse is a technique for RL guided by past learned similar policies. Policy reuse method relies on using the past policies as a probabilistic bias where the learning agent faces three choices: the exploitation of the ongoing learned policy, the exploration of random unexplored actions, and the exploitation of past policies. The key component of policy reuse is a similarity function to estimate the similarity of past policies with respect to a new one. Specifically, let W_i be the gain obtained while reusing the past policy π_i, and policy reuse method use such value to measure the usefulness of reusing the policy π_i to learn the new policy π_{new}. Policy reuse introduces a solution consists of following a softmax strategy using the values W_i and W_{new}, where the probability of using each policy can be expressed as

$$\Pr(\pi_i) = \frac{\exp(\nu W_i)}{\sum_{j=0}^{N}\exp\left(\nu W_j\right)} \tag{10}$$

where ν is a temperature parameter, and $W_{\text{new}} = W_0$ is the average reward when following the current learned policy π_{new}. This provides a way to decide whether to exploit the past policies or the new one.

Practice has shown that when a very similar policy is included in the set of policies to be reused, the improvement on learning is very high, and when the algorithm discovers that reusing the past policies is not useful, it will follow the best strategy available, which is the new policy. As such, policy reuse method in [22] can be viewed as a stochastic policy, whose actions are generated from a mixture distribution (e.g., mixture Gaussian policy) instead of a single distribution, e.g., the Gaussian policy [23].

2) Motivation for Proposed Hybrid RL Framework: In policy reuse method, the probability of using each subpolicy, i.e.,

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following each subdistribution, is determined by a heuristic method and the convergence of the overall algorithm is not guaranteed. In other words, only the parameter of the new policy is optimized based on RL. Thus, we propose a *hybrid RL framework*, which essentially can also be viewed as a stochastic policy with parameters $\boldsymbol{\theta} = [\boldsymbol{p}; \boldsymbol{\gamma}_0] \in \boldsymbol{\Theta}$ to be optimized based on the policy gradient, where $\boldsymbol{\gamma}_0$ is the parameter of the new policy and $\boldsymbol{p}$ is the probability of using each subpolicy. Such a stochastic policy can be seen as a generalization of the conventional stochastic policy with only a single subdistribution, e.g., the Gaussian policy is a special case when there is only one subpolicy/subdistribution. In addition, the hybrid RL framework contains not only old similar policies but also the DK policy to improve robustness. This is because the DK policy constructed using expert knowledge is more robust (i.e., has better generalization ability and interpretability), but usually has some gap w.r.t. the optimal solution, while old policies trained under other similar environments can provide a good initial performance to accelerate the convergence but sometimes are not robust (e.g., due to the limited representational capacity of practical DNNs and the lack of interpretability). The proposed hybrid RL framework achieves robustness as well as fast convergence by proceeding time-sharing among the ongoing learned new policy, the DK policy and old policies.

3) Details of Proposed Hybrid RL Framework: In the hybrid RL framework, there are $N \geq 1$ old policies $\pi_1, \ldots, \pi_N$ learned under other similar environments, a DK policy π_{N+1} and a new policy π_0. At each time l, the nth policy is used with probability p_n, $n \in [0, \ldots, N+1]$. We use $\boldsymbol{\gamma}_n \in \boldsymbol{\Upsilon}$ to denote the parameters of the DNN for the nth policy with $n = 0, 1, \ldots, N$. The old policies and the DK policy are fixed and the new policy together with the reuse probabilities $\boldsymbol{p} = [p_0, p_1, \ldots, p_{N+1}]^T$ will be updated.

Specifically, for the old policies and the new policy, we employ the commonly used Gaussian policy [23] with mean $\boldsymbol{\mu}_n$ and diagonal elements of the covariance matrix Σ_n parameterized by $\boldsymbol{\mu}_n = f_{m_\mu}(\boldsymbol{\gamma}_n; \boldsymbol{s}) \subseteq \mathbb{R}^{n_a}$ and $\mathrm{Diag}(\Sigma_n) = f_{m_\sigma}(\boldsymbol{\gamma}_n; \boldsymbol{s}) \subseteq \mathbb{R}^{n_a}$, respectively, and keep the nondiagonal elements of Σ_n as 0. That is,

$$\pi_n(\boldsymbol{a}|\boldsymbol{s}) \propto |\Sigma_n|^{-\frac{1}{2}} \exp\Big(-\frac{1}{2}(\boldsymbol{\mu}_n - \boldsymbol{a})^\intercal \Sigma_n^{-1}(\boldsymbol{\mu}_n - \boldsymbol{a})\Big). \quad (11)$$

On the other hand, the DK policy is a deterministic policy in which the action (the weight vector) is chosen based on the queue length, i.e.,

$$\boldsymbol{a}_l = \left\{Q_i^s(l)\right\}$$

where $Q_i^s(l)$ denotes the queue length of user i at the lth time slot, and is equal to $\sum_{l'=0}^{D_i-1} Q_{i,l-l'}$. And the greedy user scheduling algorithm is then applied to maximize the WSR based on $\boldsymbol{a}_l$. The Q-weighted greedy scheduling algorithm is shown to perform well for light to moderate traffic loading, and can provide a stable scheduling performance. To facilitate the derivation of policy reuse probability gradient, we approximate the deterministic Q-weighted policy using the Gaussian policy in (11) with the mean $\boldsymbol{\mu}_{N+1}$ given by the action from the deterministic policy, and the variance set to be a small value.

Let $\pi_{\boldsymbol{\gamma}_0} = \pi_0$ denote the new policy with parameter $\boldsymbol{\gamma}_0$. Then the hybrid policy can be expressed as

$$\pi_{\boldsymbol{\theta}} = \sum_{n=0}^{N} p_n \pi_{\boldsymbol{\gamma}_n} + p_{N+1}\pi_{N+1}. \quad (12)$$

Note that the parameters of the old policies and the DK policy are fixed, while the reusing probability $\boldsymbol{p}$ and the parameter of the new policy are updated by optimizing (8). Thus, we omit the parameters of old polices and the DK policy, and the hybrid policy is expressed as $\pi_{\boldsymbol{\theta}}$ with parameters $\boldsymbol{\theta} = [\boldsymbol{p}; \boldsymbol{\gamma}_0] \in \boldsymbol{\Theta}$.

It is a nontrivial task to solve Problem (8) under the hybrid policy $\pi_{\boldsymbol{\theta}}$. Although the hard latency constraints are embodied in the objective, avoiding the use of CMDP, the unpredictable burst traffic with multitimeslots transmissions also brings sparse rewards, which poses challenges for fast-convergent algorithm design.

4) Comparison of the Proposed Framework and the Policy Reuse Method: Both the proposed HRL-RSHLC and the policy reuse method can be viewed as stochastic policies, whose actions are generated from a mixture distribution (e.g., mixture Gaussian policy) instead of a single distribution, e.g., the Gaussian policy [23]. However, there are significant differences between the proposed hybrid RL framework and the policy reuse method.

1) *The Reusing Probability:* The probability of using each subpolicy, i.e., following each subdistribution, is determined by a heuristic method and the convergence of the overall algorithm is not guaranteed in policy reuse method. In other words, only the parameter of the new policy is optimized based on RL. While the hybrid RL framework essentially can be viewed as a stochastic policy with parameters $\boldsymbol{\theta} = [\boldsymbol{p}; \boldsymbol{\gamma}_0] \in \boldsymbol{\Theta}$ to be optimized based on the policy gradient, where $\boldsymbol{\gamma}_0$ is the parameter of the new policy and $\boldsymbol{p}$ is the probability of using each subpolicy. Unlike policy reuse method, the reusing probability is updated along with the new policy by solving the corresponding MDP problem, which is proved to converge to KKT points in this article.
2) *The Reusing Policies:* The policy reuse method only reuse old similar policies, while the hybrid RL framework contains not only old similar policies but also the DK policy to improve robustness. On one hand, the proposed framework can utilize past learned similar policies to accelerate the convergence when the new policy is not well studied initially, with rigorous convergence analysis and theoretical performance guarantee. On the other hand, it makes use of the DK policy to improve robustness to avoid any catastrophic failure of DNN-based policies even if all the DNN-based policies cannot work well (e.g., due to the limited representational capacity of practical DNNs and the lack of interpretability), making it suitable for online resource scheduling.

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IV. Proposed HRL-RSHLC Algorithm

In this section, we propose an algorithm called HRL-RSHLC to solve the Problem (8). In the hybrid RL framework, there are $N \geq 1$ old policies $\pi_1, \ldots, \pi_N$ trained under other similar environments (parameterized by DNNs), a DK policy π_{N+1}, and the new policy $\pi_0 \triangleq \pi_{\gamma_0}$ (parameterized by DNN with parameter $\boldsymbol{\gamma}_0$). In each iteration l, the agent randomly chooses policy $\pi_m, m \in [0, 1, \ldots, N, N+1]$ with probability p_m. Then the agent generates the action $\boldsymbol{a}_l$ according to π_m based on the current state $\boldsymbol{s}_l$, interacts with environment and obtains the cost $C(s_l, a_l)$, and updates the data storage ψ_l. Finally, ψ_l is used to update the hybrid policy $\pi_{\boldsymbol{\theta}}$. In the following, we elaborate on the update of the policy in detail.

A. Summary of HRL-RSHLC Algorithm

We adopt the stochastic successive convex approximation (SSCA) method to handle the stochasticity and the nonconvexity of the Problem (8). Specifically, at each iteration, the objective function $J(\boldsymbol{\theta})$ are first replaced by a convex surrogate function $J_c^l(\boldsymbol{\theta})$, which is constructed by the estimated function value $\tilde{J}^l$ and the estimated gradient $\tilde{\boldsymbol{g}}^l$. Then, we can address the original problem (8) by solving a sequence of convex surrogate optimization problems. The convex surrogate function $J_c^l(\boldsymbol{\theta})$ can be seen as a convex approximation of $J(\boldsymbol{\theta})$ based on the lth iterate $\boldsymbol{\theta}^l$, which can be formulated as:

$$J_c^l(\boldsymbol{\theta}) = \tilde{J}^l + \left(\tilde{\boldsymbol{g}}^l\right)^T \left(\boldsymbol{\theta} - \boldsymbol{\theta}^l\right) + \varsigma ||\boldsymbol{\theta} - \boldsymbol{\theta}^l||_2^2 \tag{13}$$

where $\tilde{J}^l \in \mathbb{R}$ and $\tilde{\boldsymbol{g}}^l \in \mathbb{R}^{n_{\boldsymbol{\theta}}}$ are the estimate of function value $J(\boldsymbol{\theta})$ and the estimate of gradient $\nabla J(\boldsymbol{\theta})$ at the lth iteration, and ς is a positive constant. $\tilde{J}^l$ and $\tilde{\boldsymbol{g}}^l$ are updated according to

$$\tilde{J}^l = \chi_l \bar{J}^l + (1 - \chi_l)\tilde{J}^{l-1} \tag{14}$$

$$\tilde{\boldsymbol{g}}^l = \chi_l \bar{\boldsymbol{g}}^l + (1 - \chi_l)\tilde{\boldsymbol{g}}^{l-1} \tag{15}$$

where $\{\chi_l\}$ is the update step size satisfying the Assumption 1 in Section V and $\bar{J}^l$ and $\bar{\boldsymbol{g}}^l$ are the new expressions of estimate of function value and estimate of gradient at the lth iteration, whose specific forms will be given in Section IV-A.

After replacing the objective function $J(\boldsymbol{\theta})$ with the convex surrogate function $J_c^l(\boldsymbol{\theta})$, the optimal solution $\boldsymbol{\theta}_c^l = [\boldsymbol{p}_c^l; \boldsymbol{\gamma}_{0c}^l]$ is obtained by solving the following problem:

$$\boldsymbol{\theta}_c^l = \arg\min_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} J_c^l(\boldsymbol{\theta}). \tag{16}$$

Problem (16) can be viewed as a convex approximation of the original problem, which belongs to the convex quadratic problem and the closed-form solution can be easily obtained. Then parameter of the new policy $\boldsymbol{\gamma}_0$ is updated according to

$$\boldsymbol{\gamma}_0^{l+1} = \eta_l \boldsymbol{\gamma}_{0c}^l + (1 - \eta_t)\boldsymbol{\gamma}_0^l \tag{17}$$

where $\{\eta_l\}$ is the update step size satisfying the Assumption 1 in Section V. Note that we consider the hybrid policy with $\boldsymbol{\theta} = [\boldsymbol{p}; \boldsymbol{\gamma}_0] \in \boldsymbol{\Theta}$, including the policy reuse probabilities $\boldsymbol{p} = [p_0, p_1, \ldots, p_{N+1}]^T$ satisfying $\sum_{n=0:N+1} p_n = 1$ and $0 \leq p_n \leq 1 \quad \forall n$. $\boldsymbol{p}^{l+1}$ is further projected to a convex set $\Gamma = \{\boldsymbol{x} = [x_n] \in \mathbb{R}^{N+1}: \sum_{n=0:N+1} x_n = 1, 0 \leq x_n \leq 1 \quad \forall n\}$:

$$\begin{aligned} \boldsymbol{p}^{l+1} = & \min_{\boldsymbol{x}=[x_0,\ldots,x_{N+1}]} ||(1-\eta_l)\boldsymbol{p}^l + \eta_l \boldsymbol{p}_c^l - \boldsymbol{x}||_2 \\ \text{s.t.} \quad & 0 \leq x_n \leq 1 \quad \forall n \in [0, \ldots, N+1] \\ & \sum_{n=0:N+1} x_n = 1. \end{aligned} \tag{18}$$

Moreover, in order to accelerate the convergence of HRL-RSHLC, we adopt the policy reuse method in [22] to initialize the policy reuse probabilities $\boldsymbol{p}^0$, where the policy with larger sum of rewards in several time slots will be more likely chosen.Estimation of $\bar{J}^l$ and $\bar{\boldsymbol{g}}^l$

It is known that the hard latency constraints are embodied in the HLC-ET, avoiding the use of CMDP. However, it also brings sparse rewards, that is, only when a packet is delivered successfully at one time slot, the original data size would be included in the HLC-ET, and the reward during many time slots may equal to 0 due to the multitimeslots transmission of long packets. To alleviate the issues caused by the sparsity of reward, we adopt the idea by using both stored old experiences and newly added data to estimate the value of $\bar{J}^l$ and $\bar{\boldsymbol{g}}^l$. Compared to other methods, such as reward shaping, self-supervised learning and so on, old experience reusing neither needs to deign a complicated reward shaping method, nor requires to pretrain the agent with high computational complexity, and it is shown to be an efficient method without increasing too much complexity. Moreover, the proposed algorithm is well-suited to the online learning scenario when deployed in real-world systems by reusing old experiences. Specifically, the agent stores the latest $2L$ experiences, i.e., $\psi_l = \{\boldsymbol{s}_{l-2L+1}, \boldsymbol{a}_{l-2L+1}, C(\boldsymbol{s}_{l-2L+1}, \boldsymbol{a}_{l-2L+1}), \ldots, \boldsymbol{s}_l, \boldsymbol{a}_l, C(\boldsymbol{s}_l, \boldsymbol{a}_l)\}$. At the lth iteration, the agent chooses policy $\pi_m, m \in [0, 1, \ldots, N, N+1]$ with probability p_m, and generates an action $\boldsymbol{a}_l$ based on state $\boldsymbol{s}_l$, interacts with the environment and obtain cost $C(\boldsymbol{s}_l, \boldsymbol{a}_l)$. Then $\{\boldsymbol{s}_l, \boldsymbol{a}_l, C(\boldsymbol{s}_l, \boldsymbol{a}_l)\}$ is stored in ψ_l.

We save $2L$ number of experiences because the estimation of Q-value which needs a trajectory of experiences is involved in the estimation of gradient $\bar{\boldsymbol{g}}^l$. The new expression of estimate of function value at the lth iteration $\bar{J}^l$ is obtained by the sample average method based on the data storage ε_l,

$$\bar{J}^l = \frac{1}{L} \sum_{r=1}^{2L} C(\boldsymbol{s}_{l-2L+r}, \boldsymbol{a}_{l-2L+r}). \tag{19}$$

According to the theorem of policy gradient in [23], [24], the gradient of $J(\boldsymbol{\theta})$ is

$$\nabla J(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{s} \sim \mathbf{P}_{\pi_{\boldsymbol{\theta}}}, \boldsymbol{a} \sim \pi_{\boldsymbol{\theta}}(\cdot|\boldsymbol{s})} \left[Q^{\pi_{\boldsymbol{\theta}}}(\boldsymbol{s}, \boldsymbol{a}) \nabla \log \pi_{\theta}(\boldsymbol{a}|\boldsymbol{s}) \right] \tag{20}$$

where $Q^{\pi_{\boldsymbol{\theta}}}(\boldsymbol{s}, \boldsymbol{a})$ is the Q-value function, which can be formulated as

$$\begin{aligned} & Q^{\pi_{\boldsymbol{\theta}}}(\boldsymbol{s}, \boldsymbol{a}) \\ & = \mathbb{E}_{p_s \sim \pi_{\boldsymbol{\theta}}} \left[\sum_{r=0}^{\infty} (C(\boldsymbol{s}_r, \boldsymbol{a}_r) - J(\boldsymbol{\theta})) | S_0 = \boldsymbol{s}, A_0 = \boldsymbol{a} \right]. \end{aligned} \tag{21}$$

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Algorithm 1 HRL-RSHLC Algorithm

Input: The decreasing sequences $\{\chi_l\}$ and $\{\eta_l\}$, the initial policy parameters $\boldsymbol{\theta}^0 \in \boldsymbol{\Theta}$ and first accumulate $2L$ experiences.
for $l = 0, 1, \ldots$ **do**
1. Choose policy π_l from $\{\pi_m, m \in [0, ..., N+1]\}$ with probability $P(\pi_l = \pi_m) = p_m^l$. Generate the action $\boldsymbol{a}_l$ according to π_l based on the current state $\boldsymbol{s}_l$.
2. **Environment Interaction:**
(a) Obtain the scheduling scheme by maximizing the WSR in (7) with weight vector $\boldsymbol{w}_l = \boldsymbol{a}_l$.
(b) Obtain the transmitted rate of each user $R_i(\boldsymbol{s}_l, \boldsymbol{a}_l), i \in \mathcal{K}$.
(c) Obtain cost $C(\boldsymbol{s}_l, \boldsymbol{a}_l)$.
(d) Update the environment status.
3. Update the data storage ψ_l.
4. Estimate function value and gradient by (14) and (15), respectively.
5. Update the surrogate function $J_c^l(\boldsymbol{\theta})$ via (13).
6. Solve Problem (16) to obtain $\boldsymbol{\theta}_c^l$, and update policy parameter $\boldsymbol{\theta}^{l+1}$ according to (17) and (18).
end for

The new expression of estimate of gradient at the lth iteration $\bar{\boldsymbol{g}}^l$ is also obtained by the sample average method

$$\bar{\boldsymbol{g}}^l = \frac{1}{L}\sum_{r=1}^{L} \tilde{Q}^{l-2L+r}(\boldsymbol{s}_{l-2L+r}, \boldsymbol{a}_{l-2L+r}) \nabla \log \pi_{\boldsymbol{\theta}^l}(\boldsymbol{a}_{l-2L+r}|\boldsymbol{s}_{l-2L+r}) \tag{22}$$

where

$$\tilde{Q}^{l-2L+r}(\boldsymbol{s}_{l-2L+r}, \boldsymbol{a}_{l-2L+r}) = \sum_{r'=0}^{L-1}\left(C\left(\boldsymbol{s}_{l-2L+r+r'}, \boldsymbol{a}_{l-2L+r+r'}\right) - \bar{J}^l\right) \tag{23}$$

is the estimate of Q-value starting from state $\boldsymbol{s}_{l-2L+r}$ and action $\boldsymbol{a}_{l-2L+r}$, which is obtained by using a trajectory of L experiences starting from state $\boldsymbol{s}_{l-2L+r}$ and action $\boldsymbol{a}_{l-2L+r}$. Note that we can generate not only one new experience at each iteration, but also can generate multiple new experiences, which can help to accelerate the convergence of HRL-RSHLC, and the number of new experiences at each iteration is referred to as batch size in this article.

The overall algorithm is summarized in Algorithm 1.

Note that the complexity of the proposed hybrid RL framework would not increase compared to the single-policy network, such as SCAOPO and heuristic reuse probability method, since only one policy/DNN would be chose and be updated at each iteration. In practice, we can preselect only a few old DNN-based policies according to some heuristic methods, such as the policy reuse method. Thus, N is usually a small number, e.g., 2 or 3, then the increase in the storage requirement is totally acceptable.

V. Convergence and Performance Analysis

In this section, we first present the key assumptions for convergence analysis. Based on this, we prove that the proposed HRL-RSHLC algorithm converges to a KKT point of the Problem (8).

A. Key Assumptions on Problem Structure and Algorithm Parameters

1) Assumptions of Problem (8):

Assumption 1: (Assumptions on the Problem Structure):

1) There exist constants $\lambda > 0$ and $\rho \in (0, 1)$ satisfying
$$\sup_{s\in\mathcal{S}} d_{TV}\left(\mathbf{P}(\boldsymbol{s}_l|\boldsymbol{s}_0 = \boldsymbol{s}), \mathbf{P}_{\pi_{\boldsymbol{\theta}}}\right) \le \lambda\rho^l \tag{24}$$
for all $l = 0, 1, \ldots,$ where $\mathbf{P}_{\pi_{\boldsymbol{\theta}}}$ is the stationary state distribution under policy $\pi_{\boldsymbol{\theta}}$ and $d_{TV}(a, b) = \int_{s\in\mathcal{S}} |a(\mathrm{d}\boldsymbol{s}) - b(\mathrm{d}\boldsymbol{s})|$ denotes the total-variation distance between the probability measures a and b.
2) The DNNs' parameter spaces $\boldsymbol{\Theta} \subseteq \mathbb{R}^{n_{\boldsymbol{\theta}}}$ for some positive integer $n_{\boldsymbol{\theta}}$, are compact and convex, and the outputs of DNNs are bounded.
3) The cost/reward C, the derivative and the second-order derivative of $J(\boldsymbol{\theta})$ are uniformly bounded.
4) The policy $\pi_{\boldsymbol{\theta}}$ follows Lipschitz continuity over the parameter $\boldsymbol{\theta} \in \boldsymbol{\Theta}$.
5) State space $\mathcal{S} \subseteq \mathbb{R}^{n_s}$ and action space $\mathcal{A} \subseteq \mathbb{R}^{n_{\boldsymbol{a}}}$ are both compact sets for some positive integers n_s and $n_{\boldsymbol{a}}$.

Assumption 1 Kexuan Wang-1) indicates that there exists the stationary state distribution under policy $\pi_{\boldsymbol{\theta}}$, which is independent of $\boldsymbol{s}_0$. And this assumption is a standard ergodicity assumption when considering problems without episode boundaries, see e.g., [18], [23] and [25]. Assumption 1-2) is trivial in general RL problems. Assumption 1-3) and 1-4) indicate that the Lipschitz continuity of $J(\boldsymbol{\theta})$ over parameters $\boldsymbol{\theta}$, which are always assumed in the rigorous convergence analysis of RL algorithms [25], [26], and the gradient of the policy DNN is always finite, which can also be easily satisfied. Assumption 1-5) considers a common scenario where the state and action spaces can be continuous.

2) Assumptions on Step Sizes:

Assumption 2 (Assumptions on Step Size): 1) $\chi_l \to 0$, $(1/\chi_l) \le O(l^\kappa)$ for some $\kappa \in (0, 1)$, $\sum_l \chi_l l^{-1} < \infty$, $\sum_l (\chi_l)^2 < \infty$, $\sum_l \chi_l (\log^2 l \sum_{l'=l-\log l}^{l} \eta_{l'}) < \infty$.
2) $\eta_l \to 0$, $\sum_l \eta_l = \infty$, $\sum_l (\eta_l)^2 < \infty$.
3) $\lim_{l\to\infty} (\eta_l/\chi_l) = 0$.

Note that $(1/\chi_l) \le O(l^\kappa)$ for some $\kappa \in (0, 1)$ in Assumption 1-2) is almost the same as $\sum_l \chi_l = \infty$, which is a common assumption in stochastic optimization algorithms [27]. A typical choice of $\{\chi_l\}$ and $\{\eta_l\}$ satisfying Assumption 2 is $\chi_l = l^{-\kappa_1}$ and $\eta_l = l^{-\kappa_2}$, where $\kappa_1 \in (0.5, 1)$, $\kappa_2 \in (0.5, 1]$ and $\kappa_1 < \kappa_2$.

B. Convergence of HRL-RSHLC

Based on Assumptions 1 and 2, we can first prove a lemma which indicates the asymptotic consistency of the surrogate function value $\tilde{J}^l$ and the surrogate gradient $\tilde{\boldsymbol{g}}^l$.

Lemma 1 (Asymptotic Consistency of Surrogate Functions): Suppose that Assumptions 1 and 2 are satisfied, we have

$$\lim_{l\to\infty} \left|J(\boldsymbol{\theta}_l) - \tilde{J}^l\right| = 0 \tag{25}$$

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$$\lim_{l\to\infty} \left\|\nabla J(\boldsymbol{\theta}_l) - \tilde{\boldsymbol{g}}^l\right\|_2 = 0. \tag{26}$$

Please refer to Section I of our supplementary material for the proof. Then, we consider a subsequence $\{\boldsymbol{\theta}^{l_j}\}_{j=1}^{\infty}$, which converges to a limiting point $\boldsymbol{\theta}^*$ when $j \to \infty$. There exist a converged surrogate function $\tilde{J}(\boldsymbol{\theta})$ such that

$$\lim_{j\to\infty} J_c^{l_j}(\boldsymbol{\theta}) = \tilde{J}(\boldsymbol{\theta}) \quad \forall \boldsymbol{\theta} \in \boldsymbol{\Theta} \tag{27}$$

where

$$\left|J(\boldsymbol{\theta}^*) - \tilde{J}(\boldsymbol{\theta}^*)\right| = 0$$
$$\left|\left|\nabla J(\boldsymbol{\theta}^*) - \nabla \tilde{J}(\boldsymbol{\theta}^*)\right|\right|_2 = 0.$$

Then, based on Assumptions 1 and 2, and the Lemma 1, we are ready to prove the main convergence theorem, which states that Algorithm 1 is able to converge to a KKT point of Problem (8) with an arbitrary initial point.

Theorem 1 (Global Convergence of Algorithm 1): Suppose Assumptions 1 and 2 are satisfied, and since problem (8) is an MDP without constraints, an arbitrary initial point $\boldsymbol{\theta}^0$ is feasible. Denote $\{\boldsymbol{\theta}^l\}_{l=1}^{\infty}$ as the sequence generated by Algorithm 1 with a adequately small η_0. We denote L_l as the number of data samples, which is set to $O(\log l)$. Then every limiting point $\boldsymbol{\theta}^*$ of $\{\boldsymbol{\theta}^l\}_{l=1}^{\infty}$ is a KKT point of Problem (8) almost surely.

Please refer to Section II of our supplementary material for the proof. Note that in order to achieve the rigorous convergence proof, we assume that η_0 is adequately small. Although theoretically L_l approaches infinity when $l \to \infty$, the increasing rate is on the logarithmic order and is relatively slow. In practice, we notice that Algorithm 1 can still achieve a good convergence behavior when L_l is set to a constant number.

Remark 2 (Key Difference From the Related Work): Our previous work [28] is closely related to the work in this article. However, the proposed algorithm in [28] only optimizes the new policy without introducing DK policies and other old policies, while the HRL-RSHLC optimizes both the new policy and the probabilities of reusing the old policies. The innovative design of HRL-RSHLC helps to achieve superior performance with faster convergence speed and lower interaction cost.

VI. Simulation and Performance Evaluation

A. Simulation Setup

In the simulations, we adopt the clustered delay line *B* (CDL-B) model [29] to generate the channel of users. For each user i, data packets whose lengths follow a Poisson arrival distribution with mean λ_i arrive at the start of each timeslot with probability PA. We set the constant in the surrogate problem as $\zeta = 1$, and choose the step sizes as $\chi_l = (1/l^{0.6})$, $\eta_l = (1/l^{0.7})$. We set $K = \{8, 10\}$ and $N_t = \{4, 5\}$, and the simulation parameters are shown in Table II. In simulations, we have chosen various values of $\{D_i\}$, $\{\lambda_i\}$ and PA according to the parameters reported in [30], which are typical values in burst traffic application scenarios, and the proposed algorithm works under all of those traffic conditions. We report in this section, the simulation results under two configurations: 1) $\{D_i\} = \{4, 5, 6, 7, 4, 5, 6, 7\}$ timeslots,$\{\lambda_i\} = \{22, 42, 62, 82, 22, 42, 62, 82\}$ Kbit, $PA = 0.3$ and 2) $\{D_i\} = \{4, 5, 6, 7, 4, 5, 6, 7\}$ timeslots,$\{\lambda_i\} = \{30, 50, 70, 90, 30, 50, 70, 90\}$ Kbit, $PA = 0.3$.

TABLE II
Simulation Parameters

Parameter	Value
Bandwidth	58 MHz
TX power	12 dBm
Antenna array	ULA
Carrier frequency	3GHz
Path loss	[130,150] dB
τ	1ms
Number of users	8,10
Distribution of packets	Poisson distribution

TABLE III
Key Differences of RL Algorithms

Algorithm	Reuse policies	Performance guarantee
HRL-RSHLC	✓	✓
SCAOPO	no	✓
Policy reuse method	✓	no
SAC	no	✓
PPO	no	✓

B. Reference Baselines

We choose several state-of-the-art RL-based resource allocation methodologies as baselines, to demonstrate the novelty and competitiveness of the proposed algorithm: Soft actor–critic (SAC) [31] is a state-of-art DRL algorithm that solves both discrete and continuous control problems, and uses a stochastic policy and has been widely used in resource allocation problems; SCAOPO [28] is based on a single policy and adopts the constrained stochastic successive convex approximation (CSSCA) method [27] to handle the stochasticity and the nonconvexity of MDP problem; Policy reuse method has a similar framework with HRL-RSHLC, which is a stochastic policy with mixed subdistributions, but its reusing probability is updated in a heuristic way according to the principle given in [22]; PPO is a popular RL algorithm due to its simplicity and effectiveness in handling continuous control tasks in complex environments. We choose Q-weighted greedy algorithm in [32] as a baseline as well as the DK policy for reuse, which is shown to perform well for light to moderate traffic loading and can provide a stable scheduling performance. We summarized the key differences of those RL algorithms in Table III, and the unique feature of the proposed algorithm is that it is the only RL algorithm which reuses old policies to accelerate convergence as well as has theoretical performance guarantee.

C. Simulation Results and Discussions

First, the comparisons with CMDP-based HRL-RSHLC and HRL-RSHLC which directly controls the user scheduling and

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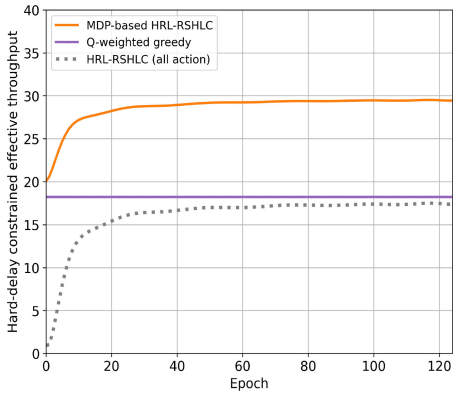


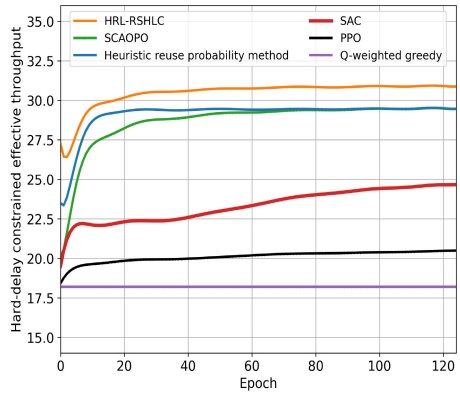
Fig. 4. Comparison with HRL-RSHLC that directly controls the user scheduling and power allocation.

power allocation are presented, to illustrate the motivation of using MDP and choosing the weight as the action. Then, we compare the performance of the proposed algorithm with other baselines under different traffic conditions. Finally, we conduct ablation experiments to better demonstrate the significance of introducing the DK policy and old policies.

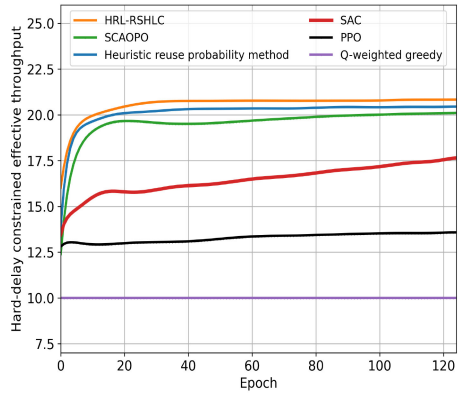
1) Comparisons With CMDP-Based HRL-RSHLC: Fig. 3(a) and (b) compare packet loss probability and HLC-ET, respectively. CMDP-based HRL-RSHLC adopts the average throughput as the reward, and the parameter ϵ is set to 0.01. Simulation results show that CMDP-based HRL-RSHLC has not fully converged within 130 epochs and has not achieved the required packet loss probability, while MDP-based HRL-RSHLC which aims to maximize the average effective throughput without an explicit constraint has achieved lower packet loss probability as well as higher effective throughput.

2) Comparisons With Other Actions: Fig. 4 compares the learning curve of HRL-RSHLC that only controls the priority weight with HRL-RSHLC that directly controls the user scheduling and power allocation, and the black stars mark the points of convergence of RL algorithms (the RL algorithm is considered converged when the reward fluctuates around a relatively stable level). It can be seen that after 130 epoch, HRL-RSHLC controlling all actions has not converged to a good solution and even performs worse than the Q-weighted greedy, which is probably due to that its enlarged action space makes it easier to get stuck in a bad local optimum. Those simulation results show that reducing the action space via only controlling the priority weight can significantly improve the convergence speed and the performance after convergence.

3) Comparisons With Baselines: We run the experiment under two traffic conditions given in Section VI-A with different user numbers and antenna numbers. Fig. 5(a) and (b) show the learning curves of HLC-ET under different traffic conditions when $K = 8$, $N_t = 4$. The proposed HRL-RSHLC converges fastest to the highest value compared to other baselines. SCAOPO converges faster than SAC and PPO, but slower than HRL-RSHLC and heuristic reuse probability method. Note that although heuristic reuse probability method achieves the second highest effective throughput under



(a)



(b)

Fig. 5. Performance comparisons for different cases when $K = 8$, $N_t = 4$. (a) Learning curve of HLC-ET under configuration (1). (b) Learning curve of hard-latency constrained effective throughput under configuration (2).

Configuration (2), it has no rigorous convergence analysis or theoretical performance guarantee for general cases.

Fig. 6(a) and (b) show the learning curves of HLC-ET under different traffic conditions when $K = 10$, $N_t = 5$. The proposed HRL-RSHLC converges fastest to the highest value compared to other baselines under both of configurations. SCAOPO converges faster than SAC, but slower than HRL-RSHLC and heuristic reuse probability method. PPO performs bad under this case, which may due to the increased state and action space.

4) Ablation Experiments: We conduct ablation experiments to better illustrate the significance of the DK policy and old policies. Fig. 7 compares the learning curve of proposed algorithm with that of HRL-RSHLC without reusing old policies, HRL-RSHLC without reusing the DK policy and HRL-RSHLC without reusing any policies. The black stars mark the points of convergence. It is clear to see that the proposed HRL-RSHLC converges to the best KKT point at a faster speed with the guidance of both old policies and the DK policy, which demonstrate the benefits of introducing both DK and old policies.

5) Performance With Imperfect CSI: In the above simulations we assume perfect CSI since the channel estimation

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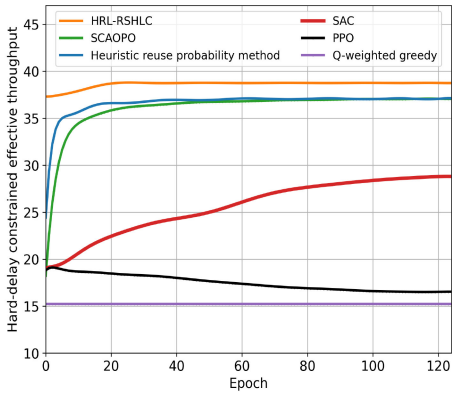
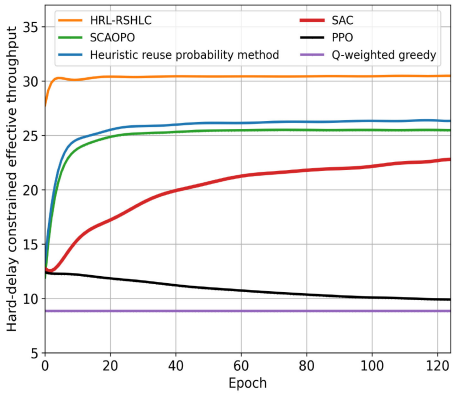


(a)



(b)

Fig. 6. Performance comparisons for different cases when $K = 10$, $N_t = 5$. (a) Learning curve of hard-latency constrained effective throughput under configuration (1). (b) Learning curve of hard-latency constrained effective throughput under configuration (2).

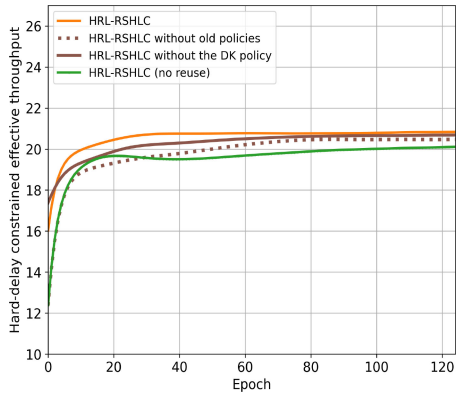


Fig. 7. Comparison with HRL-RSHLC without old policies and HRL-RSHLC without the DK policy.

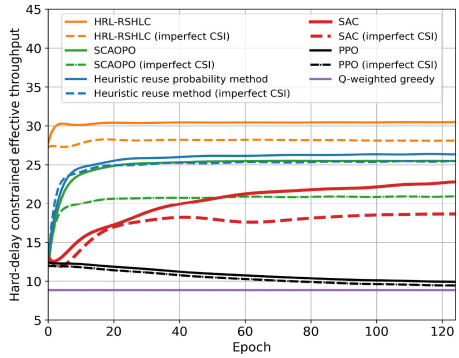


Fig. 8. Performance comparison with imperfect CSI.

process is independent with the resource scheduling process so we focus on the performance of the proposed scheduling algorithm without the added complexity of imperfect CSI. However, to demonstrate the robustness of the proposed algorithm, we further compare the performance of the algorithms with imperfect CSI. Specifically, we assume the BS performs the scheduling algorithm based on imperfect estimated CSI $\hat{\boldsymbol{H}}$, which can be expressed as

$$\hat{\boldsymbol{H}} = \boldsymbol{H} + \boldsymbol{n}_e \tag{28}$$

where $\boldsymbol{n}_e$ is the estimation noise. In the simulations, we adopted the Guassion noise to act as the channel estimation error, with normalized mean square error NMSE $= (\mathbb{E}[||\boldsymbol{H} - \hat{\boldsymbol{H}}||^2]/\mathbb{E}[||\boldsymbol{H}||^2])$ of 0.4. Fig. 8 shows the performance comparison of the algorithms with perfect CSI. The result demonstrates that although all DNN-based algorithms perform worse with imperfect CSI, the proposed HRL-RSHLC still converges faster and better than other baselines.

VII. Conclusion

We propose a novel HRL-RSHLC algorithm for resource scheduling with hard latency constraints, which reuses polices from both old policies and DK policies to improve the performance. The joint optimization of the policy reuse probabilities and new policy is formulated as an MDP, which maximizes the HLC-ET of users. In particular, the hard delay constraints are embodied in the objective function, which avoid the use of CMDP. SSCA is applied to handle the nonconvex stochastic characteristic of the MDP. We prove that the proposed HRL-RSHLC can converge to KKT points with an arbitrary initial point. Simulations show that HRL-RSHLC can achieve superior performance with faster convergence speed. However, the performance of the proposed algorithm may degrade in the nonstationary case, which is also true for most resource allocation algorithms. Future research may adopt the efficient context-aware meta-learning to address this issue.

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